

SOLA VANUA LAVA ISLAND, Vanuatu

WMO: 915510

Lat: 13.856S

Lon: 167.539E

Elev: 18

StdP: 101.11

Time Zone: 11.00 (E11)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.5	22.4	17.4	12.5	24.6	18.2	13.1	24.7	10.2	25.3	9.2	25.1	0.7	20	0.311

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.8	32.0	27.6	31.6	27.4	31.1	27.1	28.3	31.1	28.0	30.8	27.6	30.4	1.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.5	23.4	30.3	27.1	22.9	30.0	26.7	22.4	29.6	91.3	31.4	89.4	30.9	88.1	30.5	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.1	5.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	27.6	27.5	27.5	27.4	26.9	26.4	26.1	26.0	26.3	26.5	27.0	27.3
	DBStd	1.20	1.04	0.89	0.94	0.98	1.11	1.12	1.07	1.15	1.21	1.11	1.04	1.01
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6153	544	490	541	521	523	492	499	495	489	510	511	537
	CDD18.3	3111	286	257	283	271	265	242	241	237	239	252	261	279
	CDH23.3	29816	3048	2690	2910	2715	2576	2132	1982	1906	2093	2276	2599	2891
CDH26.7	7060	997	891	883	761	525	318	231	226	301	421	648	859	
Wind	WSAvg	1.7	1.1	1.3	1.0	0.8	2.0	2.1	2.8	2.7	2.5	1.8	1.4	1.1
Precipitation	PrecAvg	4078	399	343	413	431	364	276	266	231	269	351	399	343
	PrecMax	5838	701	653	962	782	676	551	517	592	553	828	884	870
	PrecMin	2435	84	121	173	184	49	46	41	35	18	11	57	54
	PrecStd	902	150	128	188	146	203	132	120	166	152	231	191	183
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	32.5	32.1	32.1	31.3	30.9	30.3	30.4	30.2	30.9	31.6	32.1
		MCWB	27.9	27.6	27.5	27.9	27.5	27.1	26.7	26.4	26.2	26.6	27.1	27.7
	2%	DB	32.0	31.8	31.5	31.3	30.5	29.7	29.2	29.3	29.5	30.1	30.7	31.5
		MCWB	27.6	27.3	27.4	27.5	26.9	26.3	25.7	25.4	25.7	26.2	26.7	27.4
	5%	DB	31.2	31.2	30.9	30.7	29.6	28.9	28.5	28.6	28.9	29.4	30.1	30.8
		MCWB	27.1	27.1	27.1	27.1	26.3	25.8	25.1	25.0	25.4	25.8	26.4	27.0
	10%	DB	30.6	30.5	30.3	30.0	28.9	28.2	27.9	27.9	28.2	28.8	29.5	30.1
		MCWB	26.8	26.7	26.8	26.8	25.8	25.3	24.7	24.5	25.0	25.4	26.0	26.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.7	28.6	28.4	28.7	28.1	27.6	27.2	27.1	27.0	27.4	28.0	29.1
		MCDB	31.9	31.7	31.1	31.1	30.6	30.1	29.6	30.0	29.4	30.0	30.6	31.2
	2%	WB	28.1	28.0	27.9	28.0	27.4	26.8	26.4	26.1	26.4	26.8	27.3	28.0
		MCDB	31.3	30.9	30.8	30.9	29.7	28.9	28.4	28.5	28.7	29.4	29.9	30.6
	5%	WB	27.6	27.5	27.5	27.5	26.9	26.4	25.9	25.6	26.0	26.3	27.0	27.5
		MCDB	30.6	30.4	30.3	30.1	29.1	28.4	27.7	27.8	28.3	28.7	29.4	30.1
	10%	WB	27.2	27.1	27.1	27.0	26.5	25.9	25.4	25.2	25.5	25.9	26.5	27.0
		MCDB	30.0	29.9	29.8	29.5	28.4	27.8	27.2	27.2	27.7	28.1	28.9	29.7
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.2	5.7	5.3	4.1	3.6	3.6	3.7	3.5	4.3	4.9	5.4
		MCDBR	7.1	7.2	6.9	6.4	5.1	4.7	4.6	4.8	4.3	5.5	5.8	6.6
		MCWBR	3.7	3.8	3.5	3.2	2.7	2.5	2.6	2.6	2.3	3.0	3.0	3.5
	5% WB	MCDBR	6.7	6.6	6.5	6.1	5.0	4.5	4.3	4.5	4.3	5.4	5.6	6.2
		MCWBR	3.8	3.8	3.6	3.3	2.8	2.6	2.8	2.7	2.5	3.2	3.2	3.8
Clear-Sky Solar Irradiance	taub	0.408	0.402	0.396	0.399	0.406	0.409	0.415	0.409	0.413	0.419	0.419	0.401	
	taud	2.549	2.569	2.597	2.584	2.551	2.534	2.502	2.520	2.504	2.497	2.508	2.562	
	Ebn at Noon	936	934	921	885	846	825	828	863	891	908	920	941	
	Edh at Noon	110	107	101	97	95	94	99	102	109	114	114	108	
All-Sky Solar Radiation	RadAvg	6.09	5.98	5.50	4.81	4.30	3.96	4.17	4.78	5.39	5.78	5.96	6.16	
	RadStd	0.61	0.60	0.53	0.43	0.41	0.25	0.23	0.35	0.43	0.60	0.48	0.51	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page